

Isaac Au

ayhisaac@gmail.com | (412) 378-7254 | <https://www.linkedin.com/in/isaacayh/>

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. in Computer Science, **GPA:** 3.9, Dean's List High Honors (2023, 2024, 2025)

Aug 2023 – May 2027

Coursework: Computer Vision (Graduate), Natural Language Processing (Graduate), Distributed Systems, Machine Learning, Computer Systems, Database Design, Application Development, Computational Finance

Teaching Assistant: Reasoning With Data (Fall 2025)

SKILLS

Programming Languages: Python, C, C++, Go (Golang), SQL, Bash/Shell, Swift

Frameworks & Tools: Docker, AWS, Azure, Git, MongoDB, PostgreSQL, FastAPI (REST APIs), Vue, Claude Code

ML/AI Libraries: PyTorch, LlamaFactory, Hugging Face, AWS Bedrock, Weights & Biases (wandb)

Languages: Fluent in English, Cantonese, Mandarin Chinese

WORK & RESEARCH EXPERIENCE

TDK

San Jose, CA & Pittsburgh, PA

Machine Learning Intern

May 2026 – Present

- Engineered internal tooling for LLM-guided evolutionary search (modeled on Google Deepmind's AlphaEvolve), automating the optimization of TinyML models across competing objectives orchestrated on AWS Bedrock with experiment tracking in Weights & Biases.
- Deployed it against a production car crash detection model for wearables, cutting model footprint 35% with no loss in rare-event F1 (+1.3%) versus the production baseline.

Ovis Medical Solutions

Hong Kong

Co-founder, Technical Lead

Feb 2024 – Feb 2026

- Designed a three-stage AI pipeline that chains LLM outputs through progressive refinement to produce severity classifications with confidence scores from unstructured user conversations.
- Collaborated with University of Hong Kong oncologists for ERB compliance and understanding patient needs.
- Architected a hybrid inference routing layer using a classifier to direct private queries to a local model (Google MedGemma) and complex reasoning to OpenAI, **cutting API costs by ~40%** and improving output quality.
- Secured \$100,000 in SAFE funding by leading a 5-person team through the technical MVP lifecycle.

Research in Medical LLM Synthetic Data and Evaluation

Pittsburgh, PA

Researcher

May 2025 – Sept 2025

- Engineered a scalable synthetic data generation pipeline utilizing an LLM-as-a-judge framework, autonomously filtering 2M+ clinical entries to curate a 50k-row high-fidelity instruction dataset.
- Orchestrated LoRA Supervised Fine-Tuning (SFT) via LlamaFactory, boosting medical reasoning accuracy by ~15% on curated MC benchmark of complex cases over foundation Qwen models to quantify CoT bottlenecks.

Rice Robotics

Hong Kong

Software Development Intern (Embodied AI)

May 2024 – Jul 2024

- Developed an Azure OpenAI autonomy system for an embodied AI robot, building a multimodal pipeline for structured motor commands and real-time emotional displays. Delivered MVP to 10 investors in 6 weeks.
- Slashed full round-trip latency by 3.5x** (from ~10s to ~3s) by multithreading the multimodal inference pipeline and parallelizing sentiment analysis and motor command generation against streamed output tokens.

PROJECTS

DropIn — Real-Time Sensor Fusion Motion Capture on Consumer Devices

Feb 2026-Present

- Engineered a real-time mocap system using iOS IMU data as a low-cost alternative to \$50k+ optical setups, building a kinematic solver with quaternions to reconstruct rider kinematics without external cameras.
- Synchronized high-throughput (100Hz) IMU streams with <50ms latency** using a custom backpressure protocol, automatically shedding stale queued frames during network congestion to guarantee state coherency.
- Designed a WebGL visualization suite with telemetry to deliver immediate, actionable technique coaching.

GUI-Grounded Generation Research

Mar 2026 – May 2026

- Built a zero-shot inference pipeline injecting frozen GUI-grounding models for computer-use agents (OmniParser v2, JEDI) as structured prompt scaffolding into a VLM code-repair model without fine-tuning.
- Drove a **+29% gain in visual fidelity** (CLIP, $p < 0.01$) on Angular, validated across 128 paired significance tests spanning 2 model sizes $\times$ 2 grounding signals $\times$ 4 frameworks on DesignBench's 111-sample repair task.